Tenrec

Session-Based Recommendations

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CSC466

The Problem

- Session-Based Recommendations are important for creating recommendations for individual sessions, without user profiles
- There is little long-term memory and interest can change quickly
- Each session is a “cold start”
- Youtube Shorts, Spotify Radio, Amazon shopping, “Up Next”

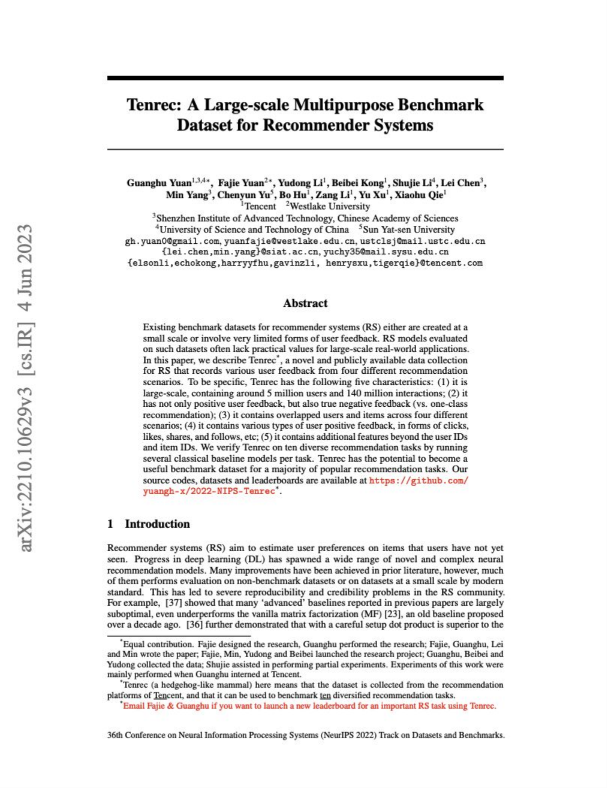
Dataset

This dataset comes from a NeurIPS paper published in 2023. It was created to benchmark various types of recommender systems.

From Tencent media platform QQ, anonymized into QB-Video, QK-Video, QB-Article, QK-Article

Real, Large, and Rich

Guanghu Yuan, Fajie Yuan, Yudong Li, Beibei Kong, Shujie Li, Lei Chen, Min Yang, Chenyun Yu, Bo Hu, Zang Li, Yu Xu, Xiaohu Qie. Tenrec: A Large-Scale Multipurpose Benchmark Dataset for Recommender Systems NeurIPS 2022.



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Tenrec: A Large-scale Multipurpose Benchmark Dataset for Recommender Systems

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Abstract

Existing benchmark datasets for recommender systems (RS) either are created at a small scale or involve very limited forms of user feedback. RS models evaluated on such datasets often lack practical values for large-scale real-world applications. In this paper, we describe Tenrec*, a novel and publicly available data collection for RS that records various user feedback from four different recommendation scenarios. To be specific, Tenrec has the following five characteristics: (1) it is large-scale, containing around 5 million users and 140 million interactions; (2) it has not only positive user feedback, but also true negative feedback (vs. one-class recommendation); (3) it contains overlapped users and items across four different scenarios; (4) it contains various types of user positive feedback, in forms of clicks, likes, shares, and follows, etc; (5) it contains additional features beyond the user IDs and item IDs. We verify Tenrec on ten diverse recommendation tasks by running several classical baseline models per task. Tenrec has the potential to become a useful benchmark dataset for a majority of popular recommendation tasks. Our source codes, datasets and leaderboards are available at https://github.com/yuangh-x/2022-NIPS-Tenrec*.

1 Introduction

Recommender systems (RS) aim to estimate user preferences on items that users have not yet seen. Progress in deep learning (DL) has spawned a wide range of novel and complex neural recommendation models. Many improvements have been achieved in prior literature, however, much of them performs evaluation on non-benchmark datasets or on datasets at a small scale by modern standard. This has led to severe reproducibility and credibility problems in the RS community. For example, [37] showed that many 'advanced' baselines reported in previous papers are largely suboptimal, even underperforms the vanilla matrix factorization (MF) [23], an old baseline proposed over a decade ago. [36] further demonstrated that with a careful setup dot product is superior to the

*Equal contribution. Fajie designed the research, Guanghu performed the research; Fajie, Guanghu, Lei and Min wrote the paper; Fajie, Min, Yudong and Beibei launched the research project; Guanghu, Beibei and Yudong collected the data; Shujie assisted in performing partial experiments. Experiments of this work were mainly performed when Guanghu interned at Tencent.

*Tenrec (a hedgehog-like mammal) here means that the dataset is collected from the recommendation platforms of Tencent, and that it can be used to benchmark ten diversified recommendation tasks.

*Email Fajie & Guanghu if you want to launch a new leaderboard for an important RS task using Tenrec.

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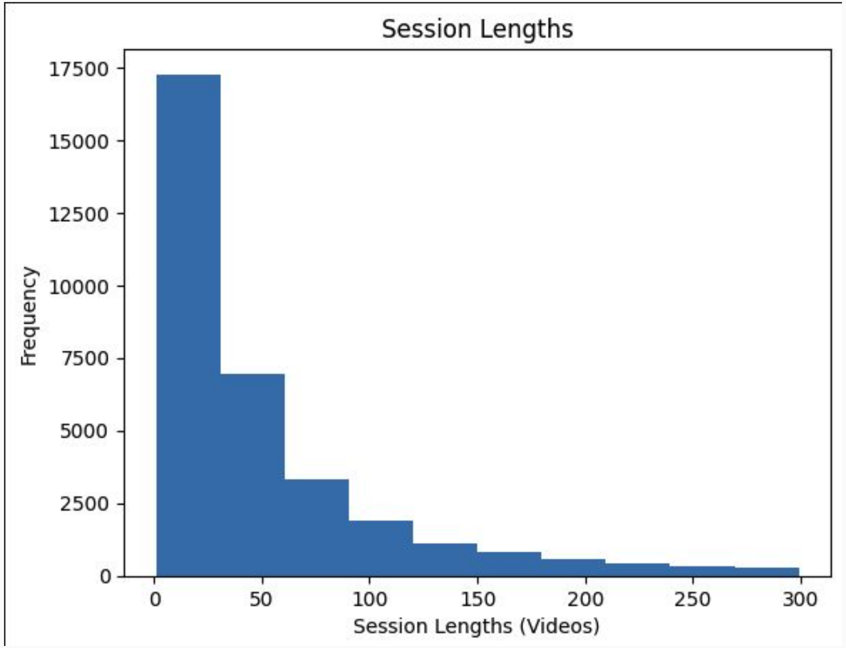
Dataset

{user_id, item_id, click, follow, like, share, video_category, watching_times, gender, age}

- No timestamps, but data is presented in sequential order
- For the smaller QK-Video, 34,000 users and 130,000 unique pieces of content
- Users were chosen to have at least five clicks
- Features are mostly self-explanatory

Sessions



The average session length is 71 videos, with 65% of users having a session length of between 0 and 50 videos

user	item	click	like
1234	85623	0	0
1234	34567	0	0
1234	76435	1	0
1234	23458	1	0
5678	98751	1	1
5678	645778	0	0
5678	34562	0	0
9876	2266	0	1
9876	64765	1	0
9876	4567456	1	0
9876	4567	0	1
9876	8975456	1	0

- We keep the session history of up to 30 videos, and mask out the most recent two videos, one for validation, one for testing

user	item	click	like
1234	85623	0	0
…	…	…	…
1234	76435	1	0
1234	23458	1	0
1234	98751	1	1
1234	645778	0	0

- Random baseline: NDCG@20 = 0.000
- Popularity Heuristic: NDCG@20 = 0.0074
  - Most clicks per video
- LSTM Recommender: NDCG@20 = 0.0304
  - History of 30 videos

Real-World Problems

- Sequences vary in length
- Huge number of items (videos, products, etc.)
- Feedback is noisy/biased and depends heavily on app design
- What should be included?
- Offline evaluation is incomplete

- Many different online recommendations can be formatted as sequences
- LSTMs handle sequence recommendations well, but attention-based mechanisms perform better
- It is difficult to determine the user experience offline
- GRU/LSTM scales well at inference time (LSTM slightly more expensive), when compared to KNN